Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

Automatic Attendance System Based on Real-Time Location and Face Recognition

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Definitions & Acronyms

No	Acronyms & Abbreviations	Definition
1	AAS	Automatic Attendance System
2	API	Application Programming Interface
3	BLE	Bluetooth Low Energy
4	PMP	Project Management Plan
5	WBS	Work Breakdown Structure
6	SRS	Software Requirements Specification
7	SDS	Software Design Specification
8	UI	User Interface
9	UX	User Experience
10	OWASP	Open Web Application Security Project

Table 3 - Definitions & Acronyms

I. Record of Changes

Date	A* M, D	In charge	Change Description
27/05	A	DungND	Added initial draft of Project Management Plan (PMP) structure.
21/06	A	DungND	Added detailed Work Breakdown Structure (WBS) and project schedule.
22/06	A	DungND	Added Communication Plan and Responsibility Assignment Matrix.
19/08	M	DungND	Modified project schedule (updated milestones and deadlines).
20/08	M	DungND	Modified Responsibility Assignment (RACI) and Communication Plan.

*A - Added M - Modified D - Deleted

Table 4 - Record of Changes

II. Project Management Plan

1. Overview

This chapter establishes the Project Management Plan for the **Automatic Attendance System (AAS)**. The plan includes the project's overall scope and effort estimation, the overall management model (Iterative & Incremental), responsibilities at a personal level, risk evaluation, and leverage. This chapter ensures the production and delivery of high-quality results at any phase of project execution.

1.1 Scope & Estimation

We categorize WBS items into three levels of complexity:

- Simple = 1 man-day
- Medium = 3 man-days
- Complex = 8 man-days

The total effort is estimated accordingly for each item.

WBS ID	WBS Item	Task Description	Complexity	Effort (days)
I	Iteration 1 – Initiation & Foundational Research (Week 1–2)			42
1.1	**Requirement Analysis**			
1.1.1	Report 1 – Introduction (outline)	Draft structure, scope, team, rationale	Medium	3
1.1.2	Report 2 – PMP (outline)	Timeline, resources, milestones, risks	Medium	3
1.1.3	Stakeholder Workshop	Goals, constraints, risk assumptions	Medium	3
1.1.4	Draft Use Case List	Initial functional use cases (no fixed count)	Medium	3
1.2	**Design**			
1.2.1	Architecture Approach	Select initial architecture for app, web, backend	Medium	3
1.3	**Coding**			
1.3.1	Backend Bootstrap	Repos, envs, CI/CD, base DB schema	Complex	8

1.3.2	BLE Research POC	Scan/broadcast prototype & limitations	Complex	8
1.3.3	Face Recognition POC	On-device FR with liveness check	Complex	8
1.4	**Testing**			
1.4.1	Pipeline Smoke Tests	CI/CD health checks & API ping	Medium	3
II	**Iteration 2 – System Design & Initial Prototyping (Week 3–4)**			**28**
2.1	**Requirement Analysis**			
2.1.1	Finalize Use Cases	Refine functional and non-functional requirements	Medium	3
2.1.2	Data Model Draft	Initial ERD, attributes, relationships	Medium	3
2.2	**Design**			
2.2.1	Low-Fidelity Wireframes	Sketch basic UI flows	Medium	2
2.2.2	System Sequence Diagrams	Capture user-system interaction	Medium	2
2.2.3	Technical Stack Validation	Validate frameworks & SDKs	Medium	2
2.3	**Coding**			
2.3.1	API Skeleton	Define endpoints, controllers	Medium	3
2.3.2	Web Portal Skeleton	Initial admin dashboard structure	Medium	3
2.3.3	Mobile App Skeleton	Initial navigation, screens	Medium	3
2.4	**Testing**			
2.4.1	Design Walkthrough	Validate mockups with peers	Simple	2
2.4.2	Prototype Demo	Present basic navigation flow	Simple	2
III	**Iteration 3 – Core Backend & API Development (Week 5–6)**			**40**
3.1	**Requirement Analysis**			

3.1.1	API Contract Workshop	Finalize request/response specs	Medium	2
3.2	**Design**			
3.2.1	DB Schema Finalization	Normalize, index, constraints	Medium	3
3.2.2	API Documentation Draft	Swagger/OpenAPI setup	Medium	2
3.3	**Coding**			
3.3.1	Authentication Service	JWT setup	Complex	6
3.3.2	User Management APIs	CRUD for user accounts	Medium	4
3.3.3	Attendance APIs	Endpoints for check-in/out	Medium	4
3.3.4	BLE Integration Service	Link with the BLE scanner	Complex	6
3.4	**Testing**			
3.4.1	Unit Test Coverage	Services and API endpoints	Medium	4
3.4.2	Postman Collection	Integration checks	Simple	3
IV	**Iteration 4 – Mobile Application Development (Week 7–8)**			**38**
4.1	**Requirement Analysis**			
4.1.1	Mobile UX Research	Validate flows with peers	Medium	2
4.1.2	Accessibility Requirements	Draft accessibility standards	Medium	2
4.2	**Design**			
4.2.1	High-Fidelity Mobile Mockups	Polished Figma screens	Medium	3
4.2.2	Mobile Navigation Flow	Final tab, drawer, deep links	Medium	2
4.3	**Coding**			
4.3.1	Login & Registration	Mobile auth screens	Medium	3
4.3.2	Attendance Check-in	BLE + Face Recognition trigger	Complex	6
4.3.3	Attendance History	List & details	Medium	3
4.4	**Testing**			

4.4.1	Emulator Testing	Android basic tests	Simple	3
4.4.2	Field Test (Pilot)	Try with real devices	Complex	5
V	**Iteration 5 – Web Portal Development (Week 9–10)**			**33**
5.1	**Requirement Analysis**			
5.1.1	Admin Needs Workshop	Reports, filtering	Medium	2
5.1.2	Security Requirements	RBAC, logging	Medium	2
5.2	**Design**			
5.2.1	Admin Dashboard Wireframes	Layout, cards, filters	Medium	2
5.2.2	Reporting Templates	Export formats	Medium	2
5.3	**Coding**			
5.3.1	Dashboard Home	Stats, charts	Medium	3
5.3.2	Attendance Logs View	Search, filter	Medium	3
5.3.3	User Management Page	List, add, deactivate	Medium	3
5.3.4	Role Management	RBAC setup	Medium	3
5.4	**Testing**			
5.4.1	Browser Testing	Chrome, Firefox, Edge	Medium	3
5.4.2	Security Testing	Pen-test basics	Complex	5
VI	**Iteration 6 – Integration & System Testing (Week 11–12)**			**35**
6.1	**Requirement Analysis**			
6.1.1	Integration Scenarios	Define cross-module flows	Medium	2
6.1.2	Test Data Requirements	Seed realistic datasets	Medium	2
6.2	**Design**			
6.2.1	Test Architecture	CI/CD automated testing plan	Medium	3
6.3	**Coding**			
6.3.1	API Gateway Setup	Central entry point	Complex	5

6.3.2	Notification Service	Email/SMS microservice	Medium	4
6.4	**Testing**			
6.4.1	System Integration Tests	End-to-end validation	Complex	6
6.4.2	Load Testing	Simulate 500 concurrent users	Complex	6
VII	**Iteration 7 – UAT & Pilot Deployment (Week 13–14)**			**32**
7.1	**Requirement Analysis**			
7.1.1	UAT Criteria	Define entry/exit conditions	Medium	2
7.2	**Design**			
7.2.1	UAT Scripts	Draft scenarios for testers	Medium	3
7.3	**Coding**			
7.3.1	Deployment Scripts	Docker manifests	Medium	3
7.3.2	Monitoring Setup	Logging & dashboards	Medium	3
7.4	**Testing**			
7.4.1	UAT Execution	Run with 10 pilot users	Complex	6
7.4.2	UAT Feedback Analysis	Summarize gaps	Medium	3
VIII	**Iteration 8 – Closure & Final Delivery (Week 15)**			**14**
8.1	**Project Closure**			
8.1.1	Prepare Final Presentation Slides	Build a defense slide deck	Simple	1
8.1.2	Demo Video	Record and edit the final demo	Medium	2
8.1.3	Rehearsal Defense	Conduct a mock defense session	Simple	1
8.1.4	Documentation Review	Final check of all deliverables	Medium	2
8.1.5	Final Project Report	Start Report 7 – Final Project Report	Complex	8

Total Estimated Effort (man-days)	***280***

Table 5 - WBS Breakdown

1.2 Project Objectives

#	Testing Stage	Test Coverage	Unit	Planned	Actual	Notes
1	Effort Usage	100% of planned man-days	Person-days	280	300	Ensure resource usage matches project estimates.
2	Reviewing	100% of documentation reviewed	No. of defects	≤10	N/A	Completeness & consistency of SRS, SDS, PMP, UI/UX artifacts.
3	Unit Test	All major functions and components	No. of defects	≤10	N/A	Verify Authentication, Settings, Device, and Session modules.
4	Integration Test	All integrated modules	No. of defects	≤15	N/A	Test mobile ↔ backend ↔ web flows, BLE/Wi-Fi data, face-ID integration.
5	System Test	100% of system functionality	No. of defects	≤20	N/A	Cover login, attendance capture, offline sync, reporting, threshold alerts.
6	Acceptance Test	Client-defined critical functionalities	No. of defects	≤20	N/A	Final stakeholder sign-off on mobile app, web portal, and analytics.

Table 6 - Project Objectives

1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	Learning curve for new technologies (BLE, Wi-Fi fingerprinting)	High	High	Organize early training & proof-of-concept in **Iteration 1–2**; assign “technology champions” to mentor peers.

2	Limited availability or unexpected absence of part-time members	High	Medium	Maintain ~20% effort buffer in each **Iteration plan**; cross-train members; rotate weekly "backup" roles.
3	Scope creep from stakeholder change requests	Mediu m	High	Apply formal change control: New requests must undergo impact analysis and be reviewed at the end of each **Iteration**.
4	Integration failures (mobile ↔ backend ↔ web portal)	High	Medium	Build integration stubs & automated checks starting **Iteration 2**; schedule dedicated "integration checkpoints" in subsequent Iterations.
5	Device compatibility issues (Android versions, BLE chipset variations)	Mediu m	Medium	Document supported device list in SRS; provide fallback manual check-in option; validate across devices in **Iteration 3–4**.
6	Security vulnerabilities (unauthorized access, data leaks)	High	Low	Conduct threat modeling in **Iteration 4**; perform code/security scans regularly; adopt OWASP Mobile Top 10 practices.
7	Performance degradation under heavy load	Mediu m	Medium	Define performance benchmarks in **Iteration 5**; perform load testing & optimize hotspots before **Iteration 7–8**.
8	Data privacy & compliance risks (location & biometric data handling)	High	Low	Implement encryption at rest/in transit; anonymize logs; review compliance with legal/IT security team in **Iteration 4–5**.

Table 7 - Project Risks

2. Management Approach

2.1 Project Process

In this project, we apply the **Iterative and Incremental** development model. The main principle of this model is to deliver the system in a sequence of iterations, where each iteration goes through the essential phases: **Requirement Analysis** → **Design** → **Coding** → **Testing**.

With this approach, the Automatic Attendance System (AAS) is gradually built and refined over multiple iterations. Lessons learned and feedback from earlier iterations are incorporated into later ones, ensuring continuous improvement and higher quality of the final product.

Each iteration delivers a working subset of features — for example:

- Iteration 1 focuses on core user authentication and basic attendance recording.

- Iteration 2 introduces BLE data capture and device management.
- Iteration 3 integrates face recognition for identity validation.
- Iteration 4 onward improves reporting, analytics, and system performance.

This approach ensures adaptability to requirement changes, risk reduction through early validation, and steady progress toward the complete system.

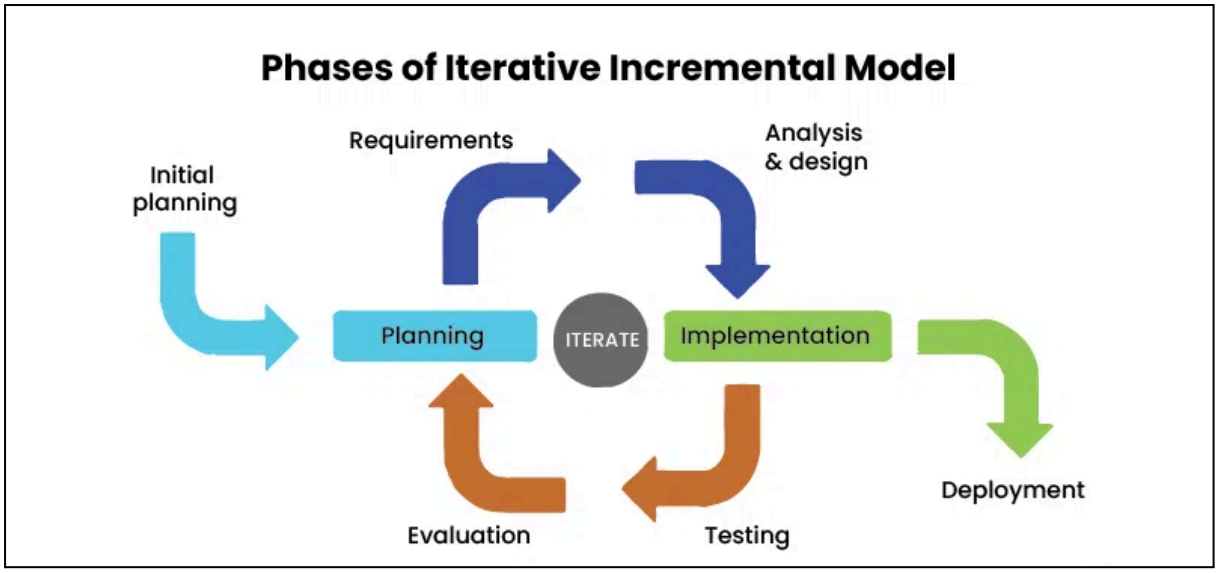


Figure 1: Development process of the iterative and incremental model

2.2 Quality Management

2.2.1 Defect Prevention

- Any defect discovered during Requirement, Design, Coding, or Testing must be immediately communicated to the responsible member.
- Defects are evaluated by severity and impact (e.g., “Does the defect affect critical attendance tracking or data accuracy?”).
- Clear deadlines for defect resolution are assigned.
- Preventive actions (such as design reviews, static code checks, and pair programming) are used to minimize future defects.

2.2.2 Coding

- Developers follow established Java/C#/React coding conventions to ensure readability and maintainability.
- Each implemented feature is peer-reviewed and accompanied by unit tests.
- Code review is mandatory before merging into the main branch to maintain system stability.
- Developers must keep local codebases updated to avoid conflicts and ensure early detection of issues.
- Security and privacy practices (e.g., encryption, OWASP guidelines) are enforced during development.

2.2.3 Reviewing

- Reviews must remain objective, focusing solely on code or documentation quality, not personal bias.
- Findings (defects, gaps, ambiguities) are logged in GitHub Project with severity, priority, and owner details.
- Assigned members are responsible for promptly addressing review findings and confirming fixes.
- Major artifacts subject to review: Software Requirement Specification (SRS), Software Design Specification (SDS), Project Management Plan (PMP), UI/UX wireframes.

2.2.4 Unit Testing

- Unit tests are written for all critical modules: Authentication, Settings, Device management, Attendance capture, and Session handling.
- Test cases must be carefully designed to cover normal, boundary, and exceptional cases.
- Results and discovered defects are logged with priority in the GitHub Project.
- The developer responsible must resolve unit test defects immediately.

2.2.5 Integration Testing

- Integration test cases validate the interaction between the mobile app, the backend API, and the web portal.
- Critical flows: BLE scanning data → backend API → attendance records update → web reporting.
- The faceid recognition module is tested with backend session validation.
- Defects found are logged with priority; responsible members must address them quickly.
- Iterations include dedicated integration checkpoints to ensure modules work smoothly together.

2.2.6 System Testing

- System-level test cases validate 100% of functional and non-functional requirements.
- Coverage includes login, attendance capture (BLE, Wi-Fi, FaceID), offline synchronization, reporting dashboards, threshold alerts, and administrative controls.
- System testing also validates compliance with performance, privacy, and security requirements.
- Defects are logged in GitHub Project, prioritized, and resolved before moving to Acceptance Testing.
- System testing ensures the solution can operate in real-world environments with multiple concurrent users and diverse device types.

Testing Method	Target
Unit Test	Validate each module in isolation (Auth, Attendance, Device, Session); ensure ≥ 95% coverage for core business logic.
Integration Test	Verify end-to-end data flows across mobile ↔ backend ↔ web; detect mismatches in BLE scanning, FaceID, and API connections.
System Test	Execute all user stories (login, check-in, offline sync, reporting); confirm the system meets both functional and non-functional requirements.
Performance Test	Ensure the system handles peak load (≥ 200 devices performing BLE/FaceID scans concurrently) with response time under 2 seconds.
Usability Test	Observe real users in classroom/event scenarios; collect feedback to refine UI/UX before final acceptance.

Table 8 - Testing Process

Defect Label	Description	Action
CRITICAL	Breaks core attendance functionality (e.g., app crash, data loss, security breach, inability to check-in).	Stop all new feature development; assign top priority and fix immediately.
MAJOR	Severely impacts key flows (e.g., authentication failures, BLE/FaceID not working, unsynchronized backend data).	Block release until resolved; address within the current iteration.
MINOR	Non-critical issues that reduce user experience (e.g., wrong error messages, slow sync, minor UI glitches).	Log in backlog; schedule fix in the next iteration based on priority.
TRIVIAL	Cosmetic or documentation issues (e.g., typos, layout misalignment, minor log inaccuracies).	Defer to future maintenance; record for reference but no immediate fix required.

Table 9 - Defect Classification

2.3 Training Plan

To ensure all team members can contribute effectively, we've scheduled targeted training on our core technologies, processes, and tools. Each session is led by the in-team expert for that topic, with mandatory attendance unless explicitly waived.

Training Area	Participants	When, Duration	Waiver Criteria
Iterative & Incremental Fundamentals	All members	Iteration 1, Day 1 (½ day)	Mandatory
Git & GitHub Workflows	All members	Iteration 1, Week 2 (1 day)	Mandatory
BLE Propagation & Wi-Fi Fingerprinting	Backend & Mobile devs	Iteration 1, Week 3 (2 days)	Mandatory
On-Device Face Recognition Integration	Mobile devs	Iteration 1, Week 3 (1 day)	Mandatory
Cross-Platform Mobile App Development	Mobile devs	Iteration 2, Week 1 (2 days)	Mandatory
React Web Portal	Web devs	Iteration 2, Week 2 (2 days)	Mandatory
Unit & Integration Testing Frameworks	All members	Iteration 3, Week 1 (1 day)	Mandatory

Table 10 - Project Training Plan

3. Project Deliverables

Deliverable	Due Date	Notes
Report 1 – Project Introduction	25/05/2025	Provides project background, objectives, scope, high-level architecture, and main use cases.
Report 2 – Project Management Plan	08/06/2025	Defines schedule, resources, risk & communication plans, and iteration breakdowns.
Project Repository & CI/CD Pipeline Setup	08/06/2025	Configured GitHub repo with automated build, test, lint, and quality checks (SonarQube, CI/CD pipeline).
Research Paper – BLE & Wi-Fi Propagation Study	22/06/2025	Summarizes pilot measurements, propagation algorithms, and fingerprinting experiments.

Report 3 – Software Requirements Specification (SRS)	22/06/2025	Functional & non-functional requirements, detailed use cases, and traceability matrix.
Report 4 – Software Design Specification (SDS)	06/07/2025	Includes architecture diagrams, data models, UI wireframes, and security design.
MVP Release v1.0 – Mobile & Web Portal	18/07/2025	First usable increment: authentication, device registration, attendance capture, and dashboard.
Incremental Release v2.0 – Enhanced Features	04/08/2025	Adds advanced reporting, BLE+FaceID hybrid attendance verification, role-based access control, and refined UI/UX.
Report 5 – Test Documentation	12/08/2025	Contains test plan, test cases, and results for unit, integration, system, performance, and usability tests.
Report 6 – User Guides & API Documentation	14/08/2025	End-user manuals and API reference with examples, workflows, screenshots, and troubleshooting.
Final Project Report & Presentation	17/08/2025	Consolidates all reports, source code, presentation slides, and lessons learned.
Project Archive & Handover Package	17/08/2025	Final GitHub release tag, deployment scripts, maintenance plan, and support SLA for transition.

Table 11 - Project Deliverables

4. Responsibility Assignments

4.1 Teams and Structures

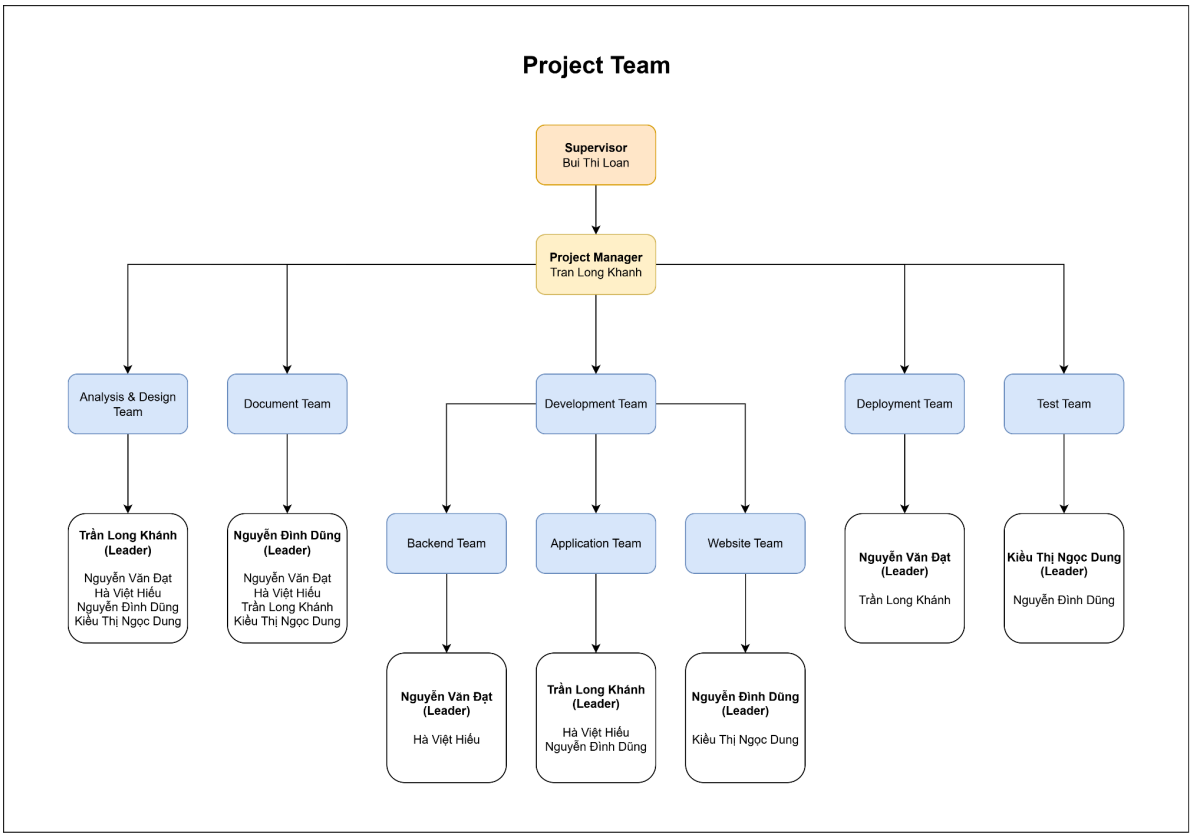


Figure 2:Team Structure

4.2 Role & Responsibilities

The table below lists the responsibilities of a person in a specific role. Each role carries different responsibilities related to the team and the assignment tasks.
(D~Do; R~Review; S~Support; I~Informed)

#	Responsibility	KhanhNT	DungND	HieuHV	DatNV	DungKTN
I. Planning						
1	Choose technology & languages	D	S	S	S	I
2	Create a project schedule	D	R	I	I	I
3	Choose a process model	D	R	I	I	I
II. Research & Experiments						

4	BLE propagation experiments & paper	D	S	S	S	S
5	Wi-Fi fingerprinting experiments & paper	S	S	D	S	S
III. Documentation & Design						
6	Report 1 – Project Introduction	R	D	I	I	I
7	Report 2 – Project Management Plan	R	D	I	I	I
8	Report 3 – Software Requirements (SRS)	R	D	S	S	D
9	Report 4 – Software Design (SDS)	R	S	D	D	I
10	UI/UX design for mobile & web	R	D	I	I	I
11	User guides & API documentation	R	D	I	D	D
IV. Development						
12	Implement backend modules (Auth, Device, Settings)	S	I	S	D	I
13	Implement mobile-app modules	D	I	S	S	I
14	Implement web-portal modules (React)	I	D	S	S	D
15	Design the database schema	R	S	D	D	I
V. Testing						
16	Test planning (unit, integration, system, UAT)	S	D	S	S	D
17	Unit testing	S	D	S	S	D

18	Integration testing	S	S	S	S	D
19	System & performance testing	S	D	S	S	D
20	Usability & acceptance testing	R	S	S	S	D
VI. Deployment & Handover						
21	MVP release & feedback collection	R	S	S	D	I
22	Final release, documentation handover & archive	R	D	S	S	I

Table 12 - Responsibility Assignment

5. Project Communications

#	Communication Item	Who / Target	Purpose	When / Frequency	Type, Tool, Method(s)
1	**Daily Meeting**	All team members	Share progress, clarify tasks, raise blockers.	Every workday at 07:00	Google Meet (15 min)
2	**Weekly Supervisor Meeting**	Team members & Instructor	Review overall progress, discuss issues, obtain feedback.	Every Tuesday at 13:00	In-class or Google Meet
3	**Iteration Planning**	Team members	Plan upcoming iteration based on WBS deliverables, allocate responsibilities.	At the start of each iteration	Google Meet (60 min)
4	**Iteration Review & Retrospective**	Team members & Instructor	Demo completed deliverables (Review); reflect on what went well and what to improve (Retro).	End of each iteration	Google Meet (Review 45 min + Retro 45 min)
5	**Backlog / Task Refinement**	Team members	Update priorities, clarify requirements, refine tasks from WBS.	Mid-iteration or as needed	GitHub Issues & Project Board

Table 13 - Project Communications

6. Configuration Management

6.1 Document Management

Our team utilizes Google Drive to organize and maintain all project documentation. This cloud service enables multiple users to work on the same files at the same time, with every change automatically saved and versioned.

Within our Drive, we structure materials into three key folders: **Documents**, which houses all formal deliverables and supplementary files; **Management**, where we keep records of team roles, workflows, and project operations; and **Testing**, dedicated to storing test plans, logs, and reports.

We author our reports and spreadsheets directly in Google Docs and Google Sheets, allowing everyone on the team to collaborate seamlessly and stay synchronized.

Link to Driver:

Tool	Activity
Google Drive	Store all documents
Googles Docs	Document Report, User Guide
Google Sheet	Project Tracking, Project Weekly, Issues Report
Canva	Slide Presentation
Figma	Design Components, Screens
Draw.io / Lucid Chart	Diagrams

Table 14 - Document Management

6.2 Source Code Management

Team members use GitHub for source code and version control following this workflow:

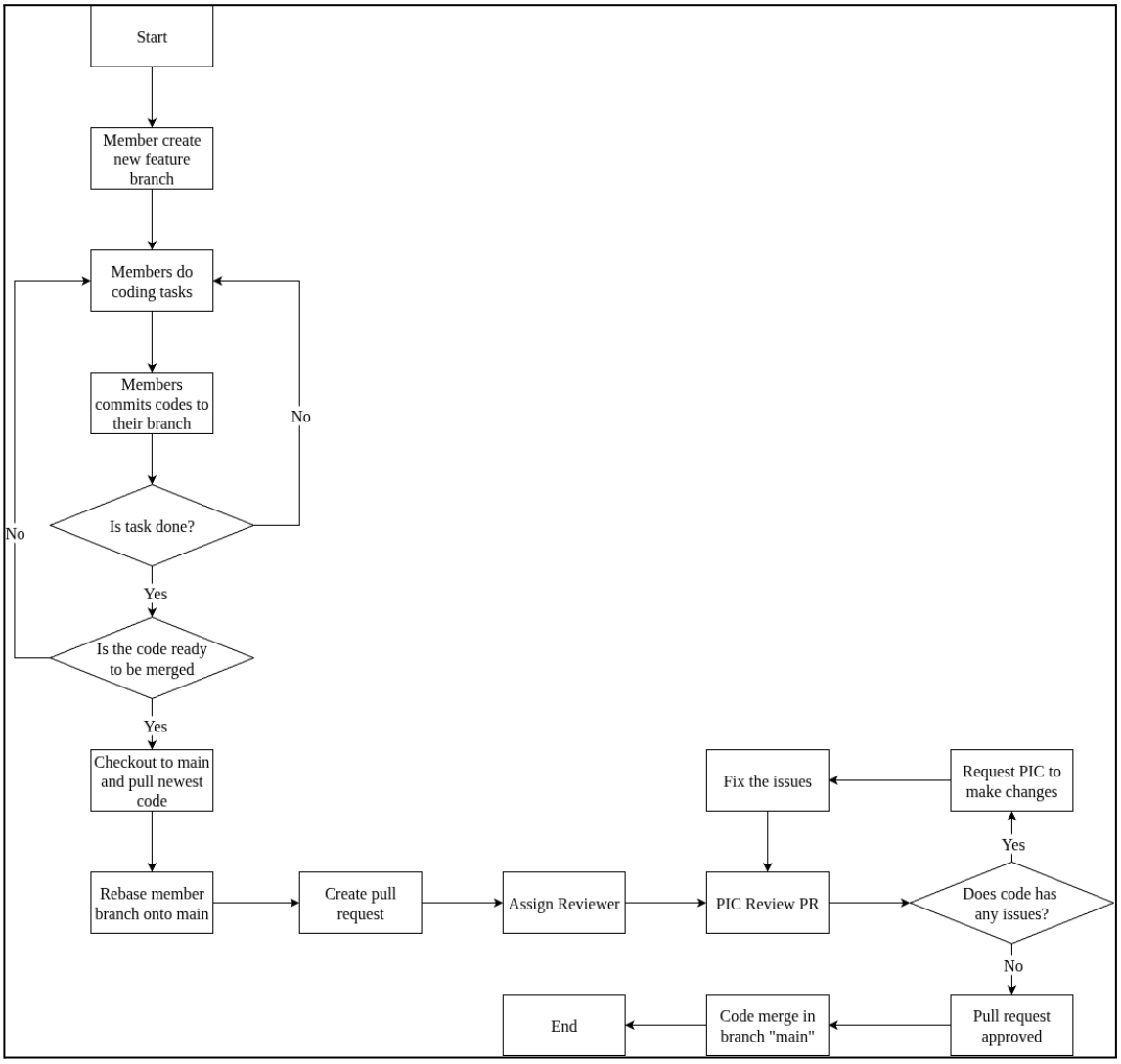


Figure 3: Source code and version control workflow

6.2.1. Convention

- All coding branches must be reviewed by team developers before merging into the main branch.
- Team members are only allowed to commit code to the local repository.
- Code must be fully tested and can run well on the local side before sending pull requests to the master repository for finished implementations.

6.2.2.Management

- Source code managed by GitHub.
- The technique leader is the person who must review carefully before merging the code of team members.
- Only team members and mentors can access the repository on GitHub.

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	Android Java, C#/ASP.NET, Javascript
Database	PostgreSQL
IDEs/Editors	Visual Studio Code, Jetbrain Rider, Android Studio
Diagramming	DrawIO, PlantUML
Documentation	Google Docs/Sheets
Version Control	Github (Source Codes), Google Drive (Documents)
Deployment server	Cloudflare Hosting

Table 15 - Tools & Infrastructures